

GUIDE K35

Datasheet

The Kalmix GUIDE K35 Series is an embedded L1/L5 RTK GNSS module family for carrier-board integration. This datasheet consolidates the published performance, interface, electrical, and package information for the K35C and K35E variants.

GNSS Performance

Published GNSS performance parameters for the K35 Series.

Parameter	Specification
Receiver type	135-channel Airoha AG3335 L1/L5 dual-band GNSS module.
Constellations and frequencies	GPS: L1C/A, L5 GLONASS: L1 Galileo: E1, E5a BeiDou: B1I, B1C, B2a QZSS: L1C/A, L5 SBAS: L1
Navigation update rate	K35C - Standard RTK: 1 Hz K35E - RTK + DR: 10 Hz
Position accuracy	Standalone: ≤ 1.0 m CEP RTK fixed: ≤ 1 cm + 1 ppm CEP
Velocity accuracy	Standalone: 0.2 m/s RTK fixed: 0.03 m/s
TTFF - AGNSS off	Cold: 26 s; Warm: 16 s; Hot: 1 s
TTFF - EASY enabled	Cold: 16 s; Warm: 2 s; Hot: 1 s
TTFF - EPO enabled	Cold: 5 s; Warm: 3 s; Hot: 1 s
Sensitivity	Acquisition: -145 dBm Reacquisition: -157 dBm Tracking: -165 dBm
Dynamic limits	Altitude: 10,000 m Velocity: 500 m/s Acceleration: 4 g

Data and Module Interfaces

GUIDE is a module-level component. The host system owns antenna selection, RF routing, power design, serial connection, and correction-data transport.

Parameter	Specification
Inertial sensor	K35E only. Integrated ST LSM6DSR 6-axis IMU for dead-reckoning sensor fusion.
Dead-reckoning support	K35E only. Supports GNSS, inertial-sensor, and wheel-tick input integration for moving-platform applications.
Timepulse output	PPS timing output with 20 ns accuracy.
Serial interfaces	3 x UART; UART0 is the primary host interface. 1 x I2C, multiplexed with UART1.
NMEA output	GGA, GSA, GSV, RMC, GLL, VTG, GST. Output rate and sentence selection are configurable.
Correction input	RTCM v3.x through the host UART data path.
RF input	50 ohm GNSS RF input; external antenna selection and bias design remain carrier-board responsibilities.

Electrical and Physical

Parameter	Specification
Power supply	3.0 V to 3.6 V DC; 3.3 V typical.
Power consumption	K35C: 25 mA during tracking/acquisition at 3.3 V. K35E: 30 mA during tracking/acquisition at 3.3 V. Backup mode: 25 uA.
Package	24-pin LCC surface-mount package.
Package dimensions	12.2 x 16.0 x 2.5 mm.
Operating temperature	-40 C to +85 C.

Integration note: verify the latest package drawing, pinout, RF-path design, external antenna requirements, and host signal-level compatibility before releasing a carrier-board schematic or PCB layout.